

CSE 5301 — Exam 2 Ultimate Study Guide

Sessions 13–23 | April 23, 2026 | Prof. Negin Fraidouni

Z-table & T-table provided | 1 A4 cheat sheet (front+back, formulas only) | Calculator needed

1. **Hypothesis Testing** (z-test, t-test, paired t-test) — **HIGH PRIORITY**
2. **Confidence Intervals / Interval Estimation** — **HIGH PRIORITY**
3. **Information Theory** (Entropy, MI, KL-Divergence) — **HIGH PRIORITY**
4. **Central Limit Theorem & Normal Distribution** — **HIGH PRIORITY**
5. **Maximum Likelihood Estimation** — **MEDIUM**
6. **Bayesian Networks** — **MEDIUM**
7. **Correlation & Linear Regression** — **MEDIUM**
8. **Statistical Inference Foundations** — **SUPPORTING**
9. **Cheat Sheet — All Formulas**
10. **Full Practice Exam**

Hypothesis Testing

HIGHEST EXAM WEIGHT

Hypothesis testing lets us make decisions about population parameters using sample data. We start with a **null hypothesis** (H_0 : the status quo) and an **alternative hypothesis** (H_1 : what we're trying to prove). We calculate a test statistic and compare it to a threshold to decide whether to reject H_0 .

The 5-Step Framework (USE THIS EVERY TIME)

1. **State hypotheses:** H_0 and H_1
2. **Choose significance level:** α (usually 0.05 or 0.01)
3. **Compute the test statistic:** z or t
4. **Find p-value OR compare to critical value**
5. **Conclusion:** Reject H_0 or Fail to reject H_0

Warning: NEVER say “accept H_0 ” — always say “fail to reject H_0 ”. This is a common point-deduction trap.

1a. Z-Test (σ known)

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

When to use: Population standard deviation σ is **KNOWN**; n is large (≥ 30) or population is normal.

Practice — Z-Test (HW6 Q3)

Problem: A sensor has known $\sigma = 0.25$ and reference mean $\mu_0 = 5$. We take 10 measurements from a new sensor: $\{4.5, 5.2, 4.3, 4.5, 4.9, 5.5, 4.2, 5.1, 4.1, 4.5\}$. Is the new sensor significantly different?

→ **Hypotheses:** $H_0 : \mu = 5$, $H_1 : \mu \neq 5$ (two-tailed)

→ **Significance:** $\alpha = 0.05$

→ **Sample mean:** $\bar{x} = 46.8/10 = 4.68$

→ **Test statistic:** $z = \frac{4.68 - 5}{0.25/\sqrt{10}} = \frac{-0.32}{0.07906} = -4.0477$

→ **p-value:** $p = 2 \times P(Z < -4.0477) \approx 0.0001$ (two-tailed)

→ **Conclusion:** $p \ll 0.05$, so **REJECT H_0** . The new sensor is significantly different.

Practice — Z-Test (HW6 Q4)

Problem: Previous studies: $\mu_0 = 1.8$ m, $\sigma = 0.3$ m. Sample of $n = 36$: $\bar{x} = 1.6839$. Has height changed? ($\alpha = 0.01$)

→ **Hypotheses:** $H_0 : \mu = 1.8$, $H_1 : \mu \neq 1.8$

→ $z = (1.6839 - 1.8)/(0.3/\sqrt{36}) = -0.1161/0.05 = -2.3222$

→ **Critical value at $\alpha = 0.01$ two-tailed:** $z_{0.005} = \pm 2.5758$

→ $|z| = 2.32 < 2.58$, and $p = 0.0202 > 0.01$

→ **Conclusion:** **FAIL TO REJECT H_0** at 1% level. Not enough evidence.

1b. One-Sample T-Test (σ unknown)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

where $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$ is the sample standard deviation.

When to use: Population σ is **UNKNOWN** → use sample s instead. Typically small sample ($n < 30$), but works for any n . Use t-table with $df = n - 1$.

Practice — One-Sample T-Test (HW6 Q1)

Problem: Algorithm runtime data ($n = 15$): {17.5, 16.3, 18.4, 14.9, 16, 16.5, 18.5, 15.9, 18, 18.5, 18.1, 15.3, 17.6, 17.7, 17.8}. Population mean $\mu_0 = 16.5$. Is there a difference?

→ **Hypotheses:** $H_0 : \mu = 16.5$, $H_1 : \mu \neq 16.5$ (two-tailed)

→ **Sample stats:** $n = 15$, $\bar{x} = 17.1333$, $s = 1.2069$

→ $t = (17.1333 - 16.5)/(1.2069/\sqrt{15}) = 0.6333/0.3116 = \mathbf{2.0323}$

→ $df = 14$, p-value (two-tailed) = 0.0615

→ **Conclusion:** $p = 0.0615 > 0.05$, so **FAIL TO REJECT H_0** . No significant difference.

Practice — One-Sided T-Test (HW6 Q5)

Problem: Method 1 scores: {10, 4, 6, 7, 12, 5, 7, 4, 12, 10, 9, 10, 8, 5, 9}. Population (Method 2) mean = 8. Does Method 1 outperform?

→ **Hypotheses:** $H_0 : \mu \leq 8$, $H_1 : \mu > 8$ (right-tailed)

→ $n = 15$, $\bar{x} = 7.8667$, $s = 2.6957$

→ $t = (7.8667 - 8)/(2.6957/\sqrt{15}) = -0.1333/0.6961 = \mathbf{-0.1916}$

→ $df = 14$, p-value (right-tailed) = 0.5746

→ **Conclusion:** $p = 0.57 > 0.05$, **FAIL TO REJECT H_0** . Method 1 does NOT outperform.

1c. Two-Sample (Welch) T-Test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

$$\text{Welch } df \approx \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

When to use: Comparing means of **two independent groups**; σ_1 and σ_2 unknown.

Practice — Two-Sample T-Test (HW6 Q2)

Problem: Body fat % — Men ($n = 13$): {13.3, 6, 20, 8, 14, 19, 18, 25, 16, 24, 15, 1, 15}; Women ($n = 10$): {22, 16, 21.7, 21, 30, 26, 12, 23.2, 28, 23}. Significant difference?

→ **Hypotheses:** $H_0 : \mu_{\text{men}} = \mu_{\text{women}}$, $H_1 : \mu_{\text{men}} \neq \mu_{\text{women}}$

→ **Men:** $\bar{x}_1 = 14.9462$, $s_1 = 6.8426$

→ **Women:** $\bar{x}_2 = 22.29$, $s_2 = 5.3197$

→ $t = (14.9462 - 22.29) / \sqrt{6.8426^2/13 + 5.3197^2/10} = -7.3438/2.5363 = -\mathbf{2.8958}$
 → Welch $df \approx 20.99$, $p = 0.0086$
 → **Conclusion:** $p = 0.0086 < 0.05$, **REJECT H_0** . Significant difference exists.

1d. Paired T-Test

Compute differences: $d_i = x_{1i} - x_{2i}$ for each pair.
 $\bar{d}$ = mean of differences, s_d = std dev of differences.

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad df = n - 1$$

When to use: Same subjects measured **TWICE** (before/after, test1/test2). Data comes in **PAIRS**.

Practice — Paired T-Test (HW6 Q6)

Problem: 16 students take 2 exams. Are the exams equally difficult?

Student	Exam 1	Exam 2	$d = E1 - E2$
Bob	63	69	-6
Nina	65	65	0
Tim	56	62	-6
Kate	100	91	9
Alonzo	88	78	10
Jose	83	87	-4
Nikhil	77	79	-2
Julia	92	88	4
Tohru	90	85	5
Michael	84	92	-8
Jean	68	69	-1
Indra	74	81	-7
Susan	87	84	3
Allen	64	75	-11
Paul	71	84	-13
Edwina	88	82	6

→ **Hypotheses:** $H_0 : \mu_d = 0$, $H_1 : \mu_d \neq 0$
 → $\bar{d} = -1.3125$, $s_d = 7.0021$, $n = 16$
 → $t = -1.3125 / (7.0021/\sqrt{16}) = -1.3125/1.7505 = -\mathbf{0.7498}$
 → $df = 15$, $p = 0.4650$
 → **Conclusion:** $p = 0.465 > 0.05$, **FAIL TO REJECT H_0** . Exams are equally difficult.

1e. Type I & Type II Errors

	H_0 is TRUE	H_0 is FALSE
Reject H_0	Type I Error (α) – False Positive	Correct! (Power = $1 - \beta$)
Fail to Reject H_0	Correct!	Type II Error (β) – False Negative

- α (**significance level**): Probability of rejecting H_0 when it's actually true. You choose this.
- β : Probability of failing to reject H_0 when it's actually false.

- **Power** = $1 - \beta$: Probability of correctly rejecting a false H_0 .

Warning: Type I = False Positive = rejecting a true H_0 . Type II = False Negative = keeping a false H_0 .
Remember: Type I = 1st kind of mistake = being too eager to reject.

1f. Z-Test for Proportions

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

where $\hat{p}$ = sample proportion, p_0 = hypothesized proportion.

When to use: Testing whether a population proportion equals a claimed value. Need $np_0 \geq 5$ and $n(1 - p_0) \geq 5$.

Practice — Z-Test for Proportions (Session 21 — Clinton Poll)

Problem: A politician claims 50% support. A poll of $n = 518$ voters finds 45% support ($\hat{p} = 0.45$). At $\alpha = 0.05$, is the true proportion different from 50%?

→ **Hypotheses:** $H_0 : p = 0.5$, $H_1 : p \neq 0.5$ (two-tailed)

$$\rightarrow z = \frac{0.45 - 0.5}{\sqrt{0.5 \times 0.5 / 518}} = \frac{-0.05}{\sqrt{0.000483}} = \frac{-0.05}{0.02198} = -2.275$$

→ **p-value (two-tailed):** $p = 2 \times P(Z < -2.275) \approx 0.023$

→ **Conclusion:** $p = 0.023 < 0.05$, so **REJECT H_0** . The true proportion significantly differs from 50%.

1g. More Worked Examples from Lecture

Practice — Z-Test (Session 20 — Blood Glucose)

Problem: Normal blood glucose: $\mu_0 = 100$ mg/dL, $\sigma = 15$. Sample of $n = 36$ patients on new diet: $\bar{x} = 108$. At $\alpha = 0.05$, does the diet change blood glucose?

→ **Hypotheses:** $H_0 : \mu = 100$, $H_1 : \mu \neq 100$

$$\rightarrow z = (108 - 100) / (15 / \sqrt{36}) = 8 / 2.5 = 3.20$$

→ **p-value:** $2 \times P(Z > 3.20) \approx 0.0014$

→ **Conclusion:** $p \ll 0.05$, **REJECT H_0** . The diet significantly changes blood glucose.

Practice — One-Sample T-Test (Session 21 — Potato Yield)

Problem: Farmer claims new fertilizer yields more than 20 tons/acre. Sample of $n = 12$: $\bar{x} = 22.1$, $s = 3.2$. At $\alpha = 0.05$, is the claim supported?

→ **Hypotheses:** $H_0 : \mu \leq 20$, $H_1 : \mu > 20$ (right-tailed)

$$\rightarrow t = (22.1 - 20) / (3.2 / \sqrt{12}) = 2.1 / 0.9238 = 2.273$$

→ $df = 11$, $t_{0.05, 11} = 1.796$ (one-tailed critical value)

→ $t = 2.273 > 1.796$

→ **Conclusion:** **REJECT H_0** . Significant evidence the fertilizer yields more than 20 tons/acre.

Decision Tree: Which test to use?

σ known? → **Z-test**

σ unknown, one sample? → **One-sample t-test**

σ unknown, two independent groups? → **Two-sample (Welch) t-test**

σ unknown, paired/matched data? → **Paired t-test**
Testing a proportion? → **Z-test for proportions**
One-sided claim (greater/less)? → **One-tailed test**
“Different from”? → **Two-tailed test**

Confidence Intervals / Interval Estimation

HIGH

Instead of estimating a parameter with a single number (point estimate), we give a range $[\hat{\theta}_l, \hat{\theta}_h]$ that likely contains the true value. The **confidence level** $(1 - \alpha)$ is the probability the interval captures the true parameter.

2a. Z-Interval (σ known, large n)

$$CI = \left[\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right]$$

$$\text{Margin of Error} = z_{\alpha/2} \cdot \sigma / \sqrt{n}$$

2b. CI with Unknown σ — Three Approaches

Approach 1: Upper Bound (Conservative)

If $\sigma \leq \sigma_{\max}$: $CI = \bar{x} \pm z_{\alpha/2} \cdot \sigma_{\max} / \sqrt{n}$ (wider but always valid)

Approach 2: Sample Standard Deviation

$S = \sqrt{\sum (x_k - \bar{x})^2 / (n - 1)}$, $CI = \bar{x} \pm z_{\alpha/2} \cdot S / \sqrt{n}$ (use t for small n)

Approach 3: Bernoulli Special Case

Conservative ($\sigma_{\max} = 0.5$): $CI = \bar{x} \pm z_{\alpha/2} / (2\sqrt{n})$

Estimated variance: $CI = \bar{x} \pm z_{\alpha/2} \cdot \sqrt{\bar{x}(1 - \bar{x}) / n}$

2c. T-Interval (σ unknown, small n)

$$CI = \left[\bar{x} - t_{\alpha/2, df} \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, df} \cdot \frac{s}{\sqrt{n}} \right], \quad df = n - 1$$

Key Insight: Common $z_{\alpha/2}$ values (MEMORIZE):

90% CI ($\alpha = 0.10$): $z_{0.05} = 1.645$ | 95% CI ($\alpha = 0.05$): $z_{0.025} = 1.96$ | 99% CI ($\alpha = 0.01$): $z_{0.005} = 2.576$

Practice — Confidence Interval

Problem: $n = 50$ light bulbs, $\bar{x} = 1200$ hours, known $\sigma = 100$. Find the 95% CI.

→ **Identify:** σ known, n large → Z-interval

→ $z_{\alpha/2} = z_{0.025} = 1.96$

→ Margin of error = $1.96 \times 100 / \sqrt{50} = 1.96 \times 14.14 = 27.72$

→ $CI = [1200 - 27.72, 1200 + 27.72] = [1172.28, 1227.72]$

→ Interpretation: We are 95% confident the true mean lifetime is between 1172.28 and 1227.72 hours.

Practice — Bernoulli CI

Problem: Poll of $n = 400$ voters, 220 support candidate A ($\bar{x} = 0.55$). Find 95% CI using (a) conservative and (b) estimated variance.

→ (a) **Conservative:** $CI = [0.55 - 1.96/(2\sqrt{400}), 0.55 + 1.96/(2\sqrt{400})] = [0.501, 0.599]$

→ (b) **Estimated:** $\hat{\sigma} = \sqrt{0.55 \times 0.45/400} = 0.02487$

→ $CI = [0.55 - 1.96 \times 0.02487, 0.55 + 1.96 \times 0.02487] = [0.501, 0.599]$

→ (In this case they're very close; the conservative method is wider in general.)

Information Theory

HIGH

Measures the amount of “surprise” or “uncertainty” in random variables. Entropy tells us how unpredictable a source is. Mutual information tells us how much knowing one variable tells us about another.

Self-Information (of a single event): $I(x) = -\log_2(P(x))$ [bits], $= -\ln(P(x))$ [nats], $= -\log_{10}(P(x))$ [Hartleys]

Entropy (average uncertainty of X):

$$H(X) = -\sum_i P(x_i) \cdot \log_2(P(x_i))$$

Joint Entropy:

$$H(X, Y) = -\sum_x \sum_y P(x, y) \cdot \log_2(P(x, y))$$

Conditional Entropy:

$$H(X|Y) = -\sum_x \sum_y P(x, y) \cdot \log_2(P(x|y)), \quad \text{where } P(x|y) = P(x, y)/P(y)$$

Mutual Information:

$$I(X; Y) = \sum_x \sum_y P(x, y) \cdot \log_2 \left[\frac{P(x, y)}{P(x) \cdot P(y)} \right]$$

Shortcuts: $I(X; Y) = H(X) + H(Y) - H(X, Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$

KL-Divergence (Relative Entropy):

$$D(P||Q) = \sum_i P(i) \cdot \log_2 \left[\frac{P(i)}{Q(i)} \right]$$

$D(P||Q) \neq D(Q||P)$ (NOT symmetric!) $D(P||Q) \geq 0$ always.

Unit Conversions: $H_{\text{Hartleys}} = H_{\text{bits}} \times 0.3010$ $H_{\text{nats}} = H_{\text{bits}} \times 0.6931$

Important Identities & Bounds

- **Chain Rule:** $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$
- **Conditional Entropy Identity:** $H(X|Y) = H(X, Y) - H(Y)$
- **Bounds on Joint Entropy:** $\max(H(X), H(Y)) \leq H(X, Y) \leq H(X) + H(Y)$
- **Mutual Information:** $I(X; Y) \geq 0$ always. $I(X; Y) = 0 \iff X$ and Y are independent.

Warning: Entropy is MAXIMIZED when all outcomes are equally likely. For n equally likely outcomes: $H = \log_2(n)$. KL-Divergence is NOT symmetric.

Practice — Full Information Theory Problem (HW4 Q1)

Problem: Given joint distribution $P(X, Y)$:

$P(X,Y)$	$X = 2$	$X = 4$	$X = 6$
$Y = 2$	0.01	0.15	0.08
$Y = 4$	0.20	0.09	0.10
$Y = 6$	0.08	0.12	0.17

Find $H(X)$:

→ **Marginals:** $P(X=2) = 0.29$, $P(X=4) = 0.36$, $P(X=6) = 0.35$

→ $H(X) = -[0.29 \log_2(0.29) + 0.36 \log_2(0.36) + 0.35 \log_2(0.35)] = \mathbf{1.5786 \text{ bits}}$

Find $H(Y)$:

→ **Marginals:** $P(Y=2) = 0.24$, $P(Y=4) = 0.39$, $P(Y=6) = 0.37$

→ $H(Y) = \mathbf{1.5547 \text{ bits}}$

Find $D(X||Y)$:

→ $D(X||Y) = 0.29 \log_2(0.29/0.24) + 0.36 \log_2(0.36/0.39) + 0.35 \log_2(0.35/0.37) = \mathbf{0.0095 \text{ bits}}$

Find $H(X,Y)$:

→ $H(X,Y) = -\sum P(x,y) \log_2(P(x,y))$ for all 9 cells = $\mathbf{2.9709 \text{ bits}}$

Find $I(X;Y)$:

→ $I(X;Y) = H(X) + H(Y) - H(X,Y) = 1.5786 + 1.5547 - 2.9709 = \mathbf{0.1624 \text{ bits}}$

→ Verify: $H(Y) - H(Y|X) = 1.5547 - 1.3923 = 0.1624 \checkmark$

Practice — Entropy of Dice (HW4 Q3)

Balanced die: Each face $P(i) = 1/6$. $H = \log_2(6) = \mathbf{2.5850 \text{ bits}}$

Loaded die: $P(1) = 0.2$, $P(2) = 0.4$, $P(3)=P(4)=P(5)=P(6) = 0.1$

→ $H = -[0.2 \log_2(0.2) + 0.4 \log_2(0.4) + 4 \times 0.1 \log_2(0.1)] = \mathbf{2.3219 \text{ bits}}$

→ Loaded die has LOWER entropy (more predictable). Makes sense!

Practice — Information Content (HW4 Q2)

Problem: Roll a fair die twice, get sequence $\{2, 3\}$. How many bits of information?

→ $P(2,3) = (1/6)(1/6) = 1/36$

→ $I = -\log_2(1/36) = \log_2(36) = 2 \log_2(6) = 2 \times 2.585 = \mathbf{5.1699 \text{ bits}}$

Practice — Unit Conversion (HW4 Q4)

Problem: $H = 3.6$ bits. Convert to Hartleys and nats.

→ Hartleys = $3.6 \times \log_{10}(2) = 3.6 \times 0.3010 = \mathbf{1.0837 \text{ Hartleys}}$

→ Nats = $3.6 \times \ln(2) = 3.6 \times 0.6931 = \mathbf{2.4953 \text{ nats}}$

Central Limit Theorem & Normal Distribution

HIGH

Central Limit Theorem (CLT)

If $X_1, X_2, \dots, X_n$ are i.i.d. random variables with mean μ and variance σ^2 , then for large n :

- The sample mean $\bar{X}$ is approximately Normal: $\bar{X} \sim N(\mu, \sigma^2/n)$
- The sum S_n is approximately Normal: $S_n \sim N(n\mu, n\sigma^2)$
- Rule of thumb: $n \geq 30$ is “large enough.”

Law of Large Numbers (LLN)

As $n \rightarrow \infty$, the sample mean $\bar{X}_n$ converges to the population mean μ .

Weak LLN: For any $\varepsilon > 0$, $P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$.

CLT tells you the *shape* of the distribution (Normal); LLN tells you the sample mean *converges* to the true mean.

Z-Score:

$$\begin{aligned} \text{Individual } X : \quad Z &= \frac{X - \mu}{\sigma} \\ \text{Sample mean } \bar{X} : \quad Z &= \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \\ \text{Sum } S_n : \quad Z &= \frac{S_n - n\mu}{\sigma\sqrt{n}} \end{aligned}$$

Normal Approximation to Binomial:

$$X \sim \text{Binomial}(n, p): \quad \mu = np, \quad \sigma = \sqrt{np(1-p)}$$

$$\text{With continuity correction: } P(X < k) \approx P\left(Z < \frac{k - 0.5 - np}{\sqrt{np(1-p)}}\right)$$

Inverse Z (Finding values from percentiles): $x = \mu + z \cdot \sigma$

Practice — CLT for Sample Mean (HW5 Q1)

Problem: $n = 100$ people, $E[X_i] = 150$ cm, $\sigma = 25$ cm. Find $P(\bar{X} < 145)$.

$$\rightarrow \text{Standard error: } \sigma_{\bar{x}} = 25/\sqrt{100} = 2.5$$

$$\rightarrow Z = (145 - 150)/2.5 = -2.0$$

$$\rightarrow P(Z < -2.0) = \mathbf{0.0228} \text{ (from Z-table)}$$

Practice — Finding Percentiles (HW5 Q2)

Problem: Heights $\sim N(65, 6^2)$. (a) Height below which shortest 30% fall; (b) height above which tallest 5% fall.

$$\rightarrow \text{(a) Need } P(X < x) = 0.30 \rightarrow z = -0.5244$$

$$\rightarrow x = 65 + (-0.5244)(6) = 65 - 3.15 = \mathbf{61.85 \text{ inches}}$$

$$\rightarrow \text{(b) Need } P(X > x) = 0.05 \rightarrow P(X < x) = 0.95 \rightarrow z = 1.6449$$

$$\rightarrow x = 65 + (1.6449)(6) = 65 + 9.87 = \mathbf{74.87 \text{ inches}}$$

Practice — Normal Approx to Binomial (HW5 Q3)

Problem: $n = 2000$ chips, failure rate $p = 0.063$. Find $P(X < 135)$.

$$\rightarrow \mu = np = 126, \sigma = \sqrt{np(1-p)} = \sqrt{118.062} = 10.8657$$

$$\rightarrow \text{With continuity correction: } Z = (134.5 - 126)/10.8657 = 0.7823$$

$$\rightarrow P(Z < 0.7823) = \mathbf{0.7830}$$

Practice — CLT for Sum (HW5 Q4)

Problem: Waiting time: $\mu = 20$ s, $\sigma = 8$ s. Find $P(900 < S_{50} < 1100)$.

$$\rightarrow E[S_{50}] = 50 \times 20 = 1000, \sigma_S = 8\sqrt{50} = 56.57$$

$$\rightarrow Z_1 = (900 - 1000)/56.57 = -1.7678$$

$$\rightarrow Z_2 = (1100 - 1000)/56.57 = 1.7678$$

$$\rightarrow P = \Phi(1.7678) - \Phi(-1.7678) = 0.9615 - 0.0385 = \mathbf{0.9229}$$

Maximum Likelihood Estimation

MEDIUM

MLE finds the parameter value θ that makes the observed data most probable. We write the likelihood function, take its log, differentiate, set to 0, and solve.

Procedure:

1. Write the likelihood: $L(\theta) = \prod P(x_i|\theta)$
2. Take the log-likelihood: $\ell(\theta) = \ln L(\theta) = \sum \ln P(x_i|\theta)$
3. Differentiate: $d\ell/d\theta = 0$
4. Solve for $\hat{\theta}$ (the MLE)
5. Verify it's a maximum (second derivative < 0)

Common MLEs:

Normal(μ, σ^2): $\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$

Bernoulli(p): $\hat{p} = \bar{x}$

Poisson(λ): $\hat{\lambda} = \bar{x}$

Exponential(λ): $\hat{\lambda} = 1/\bar{x}$

Properties: Consistent ($\hat{\theta} \rightarrow \theta$ as $n \rightarrow \infty$), asymptotically normal and efficient, invariant (if $\hat{\theta}$ is MLE of θ , then $g(\hat{\theta})$ is MLE of $g(\theta)$).

Practice — MLE for Normal Distribution

Problem: Data: $\{3, 5, 7, 9, 11\}$. Assuming Normal(μ, σ^2), find MLE for μ and σ^2 .

$\rightarrow \hat{\mu} = \bar{x} = (3 + 5 + 7 + 9 + 11)/5 = 35/5 = 7$

$\rightarrow \hat{\sigma}^2 = \frac{1}{5}[(3-7)^2 + (5-7)^2 + (7-7)^2 + (9-7)^2 + (11-7)^2] = \frac{40}{5} = 8$

Practice — MLE for Exponential

Problem: Lifetimes (hours): $\{2.1, 3.5, 1.8, 4.2, 2.9\}$. Find MLE for λ if $X \sim \text{Exp}(\lambda)$.

$\rightarrow \bar{x} = 14.5/5 = 2.9$

$\rightarrow \hat{\lambda} = 1/\bar{x} = 1/2.9 = \mathbf{0.3448}$

Bayesian Networks

MEDIUM

A Bayesian Network (BN) is a directed acyclic graph (DAG) where nodes represent random variables and edges represent probabilistic dependencies. Each node has a conditional probability table (CPT).

Joint Probability:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$$

Key Concepts:

- Parent node: directly influences child
- Root node: no parents $\rightarrow$ uses marginal $P(X)$
- Conditional independence: $X \perp Y \mid Z$ if knowing Z makes X and Y independent
- d-separation: determines conditional independence in the graph

Markov Blanket of node X :

$$\text{MB}(X) = \text{Parents}(X) \cup \text{Children}(X) \cup \text{Other parents of Children}(X)$$

Given its Markov blanket, X is conditionally independent of ALL other nodes.

Local Markov Property: Each node is conditionally independent of its non-descendants given its parents:
 $X \perp \text{NonDesc}(X) \mid \text{Parents}(X)$

Inference: $P(A \mid B=b) = P(A, B=b) / P(B=b)$

Use marginalization: $P(B=b) = \sum_A P(A, B=b)$

Practice — Bayesian Network Computation

Problem: BN: Rain $\rightarrow$ Sprinkler, Rain $\rightarrow$ WetGrass, Sprinkler $\rightarrow$ WetGrass.

Given: $P(R=T) = 0.2$, $P(S=T|R=T) = 0.01$, $P(S=T|R=F) = 0.4$, $P(W=T|S=T, R=T) = 0.99$,
 $P(W=T|S=T, R=F) = 0.90$, $P(W=T|S=F, R=T) = 0.80$, $P(W=T|S=F, R=F) = 0.0$.

Find $P(\text{Rain}=T \mid \text{WetGrass}=T)$.

$\rightarrow$ **Use Bayes:** $P(R=T|W=T) = P(W=T|R=T) \cdot P(R=T) / P(W=T)$

$\rightarrow P(W=T|R=T)$: Marginalize over S :

$\rightarrow = 0.99 \times 0.01 + 0.80 \times 0.99 = 0.0099 + 0.792 = 0.8019$

$\rightarrow P(W=T|R=F) = 0.90 \times 0.4 + 0.0 \times 0.6 = 0.36$

$\rightarrow P(W=T) = 0.8019 \times 0.2 + 0.36 \times 0.8 = 0.16038 + 0.288 = 0.44838$

$\rightarrow P(R=T|W=T) = 0.16038 / 0.44838 = \mathbf{0.3577}$

Practice — Alarm Network (Session 22)

Problem: Classic BN: Burglary(B) $\rightarrow$ Alarm(A) $\leftarrow$ Earthquake(E), Alarm $\rightarrow$ JohnCalls(J), Alarm $\rightarrow$ MaryCalls(M).

$\rightarrow$ **Joint factorization:** $P(B, E, A, J, M) = P(B) \cdot P(E) \cdot P(A|B, E) \cdot P(J|A) \cdot P(M|A)$

$\rightarrow$ **Key insight:** This is the “explaining away” example — if the alarm went off, it’s more likely to be burglary, but if we also know there was an earthquake, the probability of burglary goes down.

$\rightarrow$ **Markov blanket of B :** $\{A, E\}$ — B is independent of J, M given A .

Practice — BN: Joint Probability Computation

Problem: Chain: $A \rightarrow B \rightarrow C$. $P(A=1) = 0.3$, $P(B=1|A=1) = 0.8$, $P(B=1|A=0) = 0.2$, $P(C=1|B=1) = 0.9$, $P(C=1|B=0) = 0.1$. Find $P(A=1, B=1, C=0)$.

→ **Chain rule:** $P(A, B, C) = P(A) \cdot P(B|A) \cdot P(C|B)$

→ $P(A=1, B=1, C=0) = 0.3 \times 0.8 \times (1 - 0.9) = 0.3 \times 0.8 \times 0.1 = \mathbf{0.024}$

Correlation & Linear Regression

MEDIUM

Linear regression fits a line $\hat{y} = b_0 + b_1x$ to data points. Correlation (r) measures the strength and direction of the linear relationship between two variables.

ρ (rho) = population correlation (true but unknown). r = sample correlation (computed from data). r estimates ρ .

Pearson's r :

$$r = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{[n \sum x_i^2 - (\sum x_i)^2][n \sum y_i^2 - (\sum y_i)^2]}}$$

Also: $r = S_{xy}/(S_x \cdot S_y)$ where $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})/(n-1)$.

Properties: $-1 \leq r \leq 1$. $r = 1$: perfect positive. $r = -1$: perfect negative. $r = 0$: no linear relationship.

Regression Coefficients:

$$b_1 = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2} \quad \text{Also: } b_1 = r \cdot (S_y/S_x)$$

$$b_0 = \bar{y} - b_1 \bar{x} \quad \hat{y} = b_0 + b_1 x$$

Coefficient of Determination: $R^2 = r^2$ = proportion of variance in Y explained by X .

Residual: $e_i = y_i - \hat{y}_i$ (actual minus predicted)

Sum of Squares:

$$\text{SST} = \sum (y_i - \bar{y})^2 \text{ (Total)} \quad \text{SSR} = \sum (\hat{y}_i - \bar{y})^2 \text{ (Regression)} \quad \text{SSE} = \sum (y_i - \hat{y}_i)^2 \text{ (Error)}$$

$$\text{SST} = \text{SSR} + \text{SSE} \quad R^2 = \text{SSR}/\text{SST} = 1 - \text{SSE}/\text{SST}$$

Practice — Linear Regression

Problem: Data: $x = \{1, 2, 3, 4, 5\}$, $y = \{2, 4, 5, 4, 5\}$. Find the regression line.

→ **Sums:** $n = 5$, $\sum x = 15$, $\sum y = 20$, $\sum xy = 67$, $\sum x^2 = 55$

→ $\bar{x} = 3$, $\bar{y} = 4$

→ $b_1 = (5 \times 67 - 15 \times 20)/(5 \times 55 - 15^2) = (335 - 300)/(275 - 225) = 35/50 = 0.7$

→ $b_0 = 4 - 0.7 \times 3 = 4 - 2.1 = 1.9$

→ Regression line: **$= 1.9 + 0.7x$**

Practice — Correlation Interpretation (Session 23)

Problem: E-commerce study finds $r = 0.98$ between ad spending (x) and revenue (y). If $S_x = 200$, $S_y = 5000$, find b_1 .

→ $r = 0.98$: Very strong positive linear relationship.

→ $R^2 = 0.98^2 = 0.9604$: 96.04% of variance in revenue explained by ad spending.

→ **Slope shortcut:** $b_1 = r \times (S_y/S_x) = 0.98 \times (5000/200) = 0.98 \times 25 = \mathbf{24.5}$

→ Interpretation: For every \$1 increase in ad spending, revenue increases by $\approx \$24.50$.

Warning: Correlation $\neq$ Causation! High r only means a linear association exists, not that x causes y . Also, r only captures *linear* relationships — two variables could have a strong nonlinear relationship with $r \approx 0$.

Statistical Inference Foundations

SUPPORTING

- **Population:** The entire group we want to study.
- **Sample:** A subset of the population we actually observe.
- **Parameter:** A number describing the population (μ, σ, p). UNKNOWN.
- **Statistic:** A number computed from the sample ($\bar{x}, s, \hat{p}$). KNOWN.
- **Point Estimator:** A single value estimate for a parameter (e.g., $\bar{x}$ for μ).
- **Unbiased Estimator:** $E[\hat{\theta}] = \theta$ (on average, hits the true value).
- **Bias:** $B(\hat{\theta}) = E[\hat{\theta}] - \theta$
- **Mean Squared Error:** $MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$

If unbiased (Bias=0), then $MSE = \text{Var}$. The MLE $\hat{\sigma}^2 = (1/n) \sum (x_i - \bar{x})^2$ is biased; the sample variance $S^2 = (1/(n-1)) \sum (x_i - \bar{x})^2$ is unbiased.

Frequentist vs. Bayesian

- **Frequentist:** Parameters are fixed but unknown constants. Probability = long-run frequency. Uses MLE, confidence intervals.
- **Bayesian:** Parameters are random variables with prior distributions. Updates beliefs using Bayes' theorem. Uses posterior distributions, credible intervals.

Order Statistics

If $X_1, \dots, X_n$ are i.i.d. with CDF $F(x)$ and PDF $f(x)$, the k -th order statistic $X_{(k)}$ is the k -th smallest value.

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} \cdot [F(x)]^{k-1} \cdot [1-F(x)]^{n-k} \cdot f(x)$$

Special cases: $X_{(1)} = \min, X_{(n)} = \max$.

$$f_{(1)}(x) = n[1-F(x)]^{n-1}f(x) \quad f_{(n)}(x) = n[F(x)]^{n-1}f(x)$$

$$\text{Sample mean: } \bar{x} = \frac{1}{n} \sum x_i \quad \text{Sample variance: } S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$$

$$\text{Properties: } E[\bar{X}] = \mu \text{ (unbiased), } \text{Var}(\bar{X}) = \sigma^2/n, \quad E[S^2] = \sigma^2 \text{ (unbiased)}$$

A4 CHEAT SHEET — All Formulas

Print this section on ONE A4 sheet (front & back). No solved questions — formulas only!

Z-Score

$$Z = (X - \mu)/\sigma \quad | \quad Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \quad | \quad Z = (S_n - n\mu)/(\sigma\sqrt{n})$$

Confidence Intervals

Known σ : $CI = \bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$

Unknown σ : $CI = \bar{x} \pm z_{\alpha/2} \cdot S/\sqrt{n}$ (large n) or $\bar{x} \pm t_{\alpha/2, n-1} \cdot S/\sqrt{n}$ (small n)

Bernoulli: $CI = \bar{x} \pm z_{\alpha/2} \sqrt{\bar{x}(1-\bar{x})/n}$ Conservative: $\bar{x} \pm z_{\alpha/2}/(2\sqrt{n})$

$z_{0.05} = 1.645$ (90%) $z_{0.025} = 1.96$ (95%) $z_{0.005} = 2.576$ (99%)

Hypothesis Testing

Z-test: $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ [σ known]

T-test: $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$, $df = n-1$ [σ unknown]

Two-sample: $t = (\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$

Paired: $t = \bar{d}/(s_d/\sqrt{n})$, $df = n-1$

Proportion: $z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$

Two-tailed: reject if $|\text{test stat}| > \text{critical value}$ OR $p < \alpha$

Right-tailed: reject if test stat $>$ critical value Left-tailed: reject if test stat $<$ $-\text{critical}$

Type I error (α): reject true H_0 Type II error (β): keep false H_0

Sample Variance

$$S^2 = \sum (x_i - \bar{x})^2 / (n-1) = (\sum x_i^2 - n\bar{x}^2) / (n-1)$$

Binomial Approximation (CLT)

$X \sim \text{Bin}(n, p)$: $\mu = np$, $\sigma = \sqrt{np(1-p)}$

$$P(X < k) \approx \Phi\left(\frac{k - 0.5 - np}{\sqrt{np(1-p)}}\right) \quad [\text{continuity correction}]$$

Information Theory

Self-info: $I(x) = -\log_2(P(x))$ Entropy: $H(X) = -\sum P(x) \log_2(P(x))$

Joint: $H(X, Y) = -\sum P(x, y) \log_2(P(x, y))$ Cond: $H(X|Y) = -\sum P(x, y) \log_2(P(x|y))$

MI: $I(X; Y) = H(X) + H(Y) - H(X, Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$

KL: $D(P\|Q) = \sum P(i) \log_2(P(i)/Q(i))$ [NOT symmetric, ≥ 0]

Chain rule: $H(X, Y) = H(X) + H(Y|X)$ Bounds: $\max(H(X), H(Y)) \leq H(X, Y) \leq H(X) + H(Y)$

$I(X; Y) \geq 0$, $I(X; Y) = 0 \iff$ independent Units: bits $\rightarrow$ Hartleys $\times 0.3010$; bits $\rightarrow$ nats $\times 0.6931$

MLE Common Results

Normal: $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = (1/n) \sum (x_i - \bar{x})^2$ Bernoulli: $\hat{p} = \bar{x}$ Poisson: $\hat{\lambda} = \bar{x}$ Exp: $\hat{\lambda} = 1/\bar{x}$

Procedure: $L(\theta) = \prod P(x_i|\theta) \rightarrow \ell(\theta) = \sum \ln P(x_i|\theta) \rightarrow d\ell/d\theta = 0 \rightarrow \text{solve}$

Linear Regression

$b_1 = (n \sum xy - \sum x \sum y) / (n \sum x^2 - (\sum x)^2)$ Also: $b_1 = r \cdot (S_y/S_x)$

$b_0 = \bar{y} - b_1 \bar{x}$ $\hat{y} = b_0 + b_1 x$

$r = (n \sum xy - \sum x \sum y) / \sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}$

$R^2 = r^2$ SST = SSR + SSE $R^2 = 1 - \text{SSE}/\text{SST}$

Bayesian Networks

Joint: $P(X_1, \dots, X_n) = \prod P(X_i | \text{Parents}(X_i))$ Bayes: $P(A|B) = P(B|A)P(A)/P(B)$

Marginalization: $P(B) = \sum_A P(A, B)$

Markov blanket(X) = Parents $\cup$ Children $\cup$ CoParents Local Markov: $X \perp \text{NonDesc}(X) | \text{Parents}(X)$

Decision: Which Test?

σ known $\rightarrow$ Z-test σ unknown, 1 sample $\rightarrow$ t-test ($df = n-1$) 2 indep groups $\rightarrow$ Welch t
 Paired data $\rightarrow$ Paired t ($df = n-1$) Proportion $\rightarrow z = (\hat{p} - p_0) / \sqrt{p_0(1-p_0)/n}$

Order Statistics

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1} (1-F(x))^{n-k} f(x)$$

$$\text{Min: } f_{(1)} = n(1-F)^{n-1}f \quad \text{Max: } f_{(n)} = nF^{n-1}f$$

MSE & Bias

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2 \quad \text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta \quad \text{LLN: } \bar{X}_n \rightarrow \mu \text{ as } n \rightarrow \infty$$

Full Practice Exam

These mirror the format, difficulty, and point distribution of HW5 & HW6. The professor said “questions will be similar to examples and homework.”

Question 1 — Information Theory (20 points)

A communication channel transmits symbols A, B, C with probabilities $P(A) = 0.5$, $P(B) = 0.3$, $P(C) = 0.2$.

- Find the entropy H of this source. (7 pts)
- Find the information content of receiving symbol C. (3 pts)
- A second source transmits the same symbols with equal probability. Which source has higher entropy? Explain. (5 pts)
- Convert your answer from (a) to nats. (5 pts)

Solution — Question 1

- (a) $H = -[0.5 \log_2(0.5) + 0.3 \log_2(0.3) + 0.2 \log_2(0.2)]$
 → $= -[0.5(-1) + 0.3(-1.7370) + 0.2(-2.3219)] = -(-1.4855) = \mathbf{1.4855 \text{ bits}}$
 → (b) $I(C) = -\log_2(0.2) = \log_2(5) = \mathbf{2.3219 \text{ bits}}$
 → (c) Equal probability source: $H = \log_2(3) = 1.585 \text{ bits}$. **The equal-probability source has HIGHER entropy** because entropy is maximized when all outcomes are equally likely.
 → (d) $H_{\text{nats}} = 1.4855 \times \ln(2) = 1.4855 \times 0.6931 = \mathbf{1.0296 \text{ nats}}$

Question 2 — CLT / Normal Distribution (20 points)

The download speed of an internet service has mean $\mu = 50 \text{ Mbps}$ and $\sigma = 8 \text{ Mbps}$.

- For a sample of $n = 64$ users, find the probability that the average download speed is less than 48 Mbps. (10 pts)
- Find the download speed below which the slowest 10% of individual users fall. (10 pts)

Solution — Question 2

- (a) $\sigma_{\bar{x}} = 8/\sqrt{64} = 1.0$, $Z = (48 - 50)/1.0 = -2.0$
 → $P(\bar{X} < 48) = P(Z < -2.0) = \mathbf{0.0228}$
 → (b) Need $P(X < x) = 0.10 \rightarrow z = -1.2816$
 → $x = 50 + (-1.2816)(8) = 50 - 10.25 = \mathbf{39.75 \text{ Mbps}}$

Question 3 — Confidence Interval (15 points)

A factory produces bolts with unknown mean length. A sample of $n = 25$ bolts gives $\bar{x} = 10.2 \text{ cm}$ and $s = 0.5 \text{ cm}$. Assuming normal population:

- Find the 95% confidence interval using the t-distribution. (10 pts)
- How would the interval change if $n = 100$ instead? (5 pts)

Solution — Question 3

- (a) $df = 24$, $t_{0.025, 24} \approx 2.064$
- $ME = 2.064 \times 0.5 / \sqrt{25} = 2.064 \times 0.1 = 0.2064$
- $CI = [10.2 - 0.2064, 10.2 + 0.2064] = [9.994, 10.406]$
- (b) With $n = 100$, use z (large n). $z_{0.025} = 1.96$, $ME = 1.96 \times 0.5 / \sqrt{100} = 0.098$
- $CI = [10.102, 10.298]$ — **Much narrower!** Larger $n \rightarrow$ more precise.

Question 4 — Z-Test (15 points)

A manufacturer claims average battery life is $\mu_0 = 500$ hours with known $\sigma = 30$ hours. A consumer group tests $n = 36$ batteries and finds $\bar{x} = 490$ hours. At $\alpha = 0.05$, is the battery life less than claimed?

Solution — Question 4

- **Hypotheses:** $H_0 : \mu = 500$, $H_1 : \mu < 500$ (LEFT-tailed)
- $z = (490 - 500) / (30 / \sqrt{36}) = -10 / 5 = -2.0$
- **Critical value (left-tailed, $\alpha = 0.05$):** $z_{0.05} = -1.645$
- $|z| = 2.0 > 1.645$, and $p = P(Z < -2.0) = 0.0228 < 0.05$
- **Conclusion:** **REJECT H_0** . Significant evidence that battery life is less than 500 hours.

Question 5 — Paired T-Test (15 points)

A diet program measures weight of 8 people before and after:

Person	Before	After
1	185	178
2	192	190
3	206	195
4	177	177
5	225	210
6	168	165
7	245	233
8	198	195

At $\alpha = 0.05$, did the diet produce significant weight loss?

Solution — Question 5

- **Differences $d = \text{Before} - \text{After}$:** $\{7, 2, 11, 0, 15, 3, 12, 3\}$
- $\bar{d} = 53 / 8 = 6.625$
- $s_d = \sqrt{209.875 / 7} = \sqrt{29.982} = 5.4756$
- **Hypotheses:** $H_0 : \mu_d = 0$, $H_1 : \mu_d > 0$ (right-tailed)
- $t = 6.625 / (5.4756 / \sqrt{8}) = 6.625 / 1.9362 = 3.4213$
- $df = 7$, $t_{0.05, 7} = 1.895$ (one-tailed critical value)
- $t = 3.42 > 1.895$
- **Conclusion:** **REJECT H_0** . The diet produced statistically significant weight loss.

Question 6 — Linear Regression (15 points)

Given data: $x = \{2, 4, 6, 8, 10\}$, $y = \{3, 7, 8, 12, 14\}$.

- (a) Find the regression line $\hat{y} = b_0 + b_1x$. (10 pts)

(b) Predict $\hat{y}$ when $x = 7$. (2 pts)

(c) Compute R^2 . (3 pts)

Solution — Question 6

→ **Sums:** $n = 5$, $\sum x = 30$, $\sum y = 44$, $\sum xy = 316$, $\sum x^2 = 220$, $\sum y^2 = 462$

→ $\bar{x} = 6$, $\bar{y} = 8.8$

→ **(a)** $b_1 = (5 \times 316 - 30 \times 44) / (5 \times 220 - 30^2) = (1580 - 1320) / (1100 - 900) = 260 / 200 = 1.3$

→ $b_0 = 8.8 - 1.3 \times 6 = 1.0$

→ Regression line: **$= 1.0 + 1.3x$**

→ **(b)** $\hat{y}(7) = 1.0 + 1.3 \times 7 = 1.0 + 9.1 = \mathbf{10.1}$

→ **(c)** $r = 260 / \sqrt{200 \times 414} = 260 / \sqrt{82800} = 260 / 287.75 = 0.9036$

→ $R^2 = 0.9036^2 = \mathbf{0.8165} \rightarrow 81.65\%$ of variance in y is explained by x .

EXAM STRATEGY REMINDERS

1. Partial credit exists — ALWAYS show your work, write formulas, show steps.
2. Start with the question you know best.
3. Write down the hypotheses even if you can't finish the calculation.
4. If stuck: write the formula, plug in what you know, state what you'd do next.
5. Check your cheat sheet for the right formula FIRST.
6. Z and T tables are PROVIDED — don't waste cheat sheet space on them.
7. Double-check: "fail to reject" NOT "accept."
8. Time: ~80 minutes — budget ~12 min per question.